

CURRENT MIRROR

Deva Prassad S | Roll No: 2025BEC0019

Aim

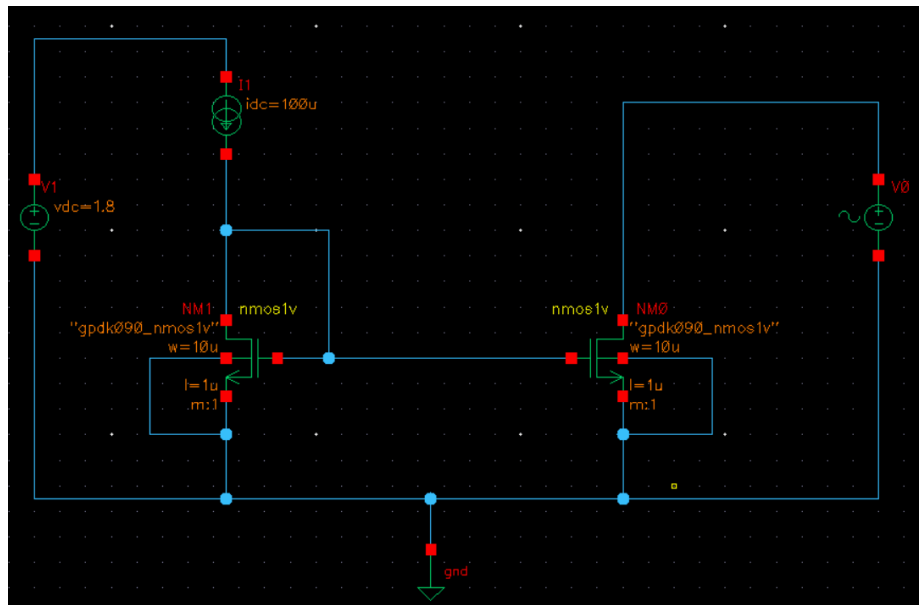
To design a simple two-transistor NMOS current mirror in GPDK090 technology and study its DC transfer characteristics and small-signal AC output resistance.

Design Specifications

Parameter	Value
Technology	GPDK090 (90 nm CMOS)
I _{ref}	100 μ A
VDD	1.8 V
W (NM0, NM1)	10 μ m
L (NM0, NM1)	1 μ m

Circuit Description

The circuit consists of two identical NMOS transistors, NM1 and NM0. NM1 is diode-connected (gate shorted to drain) and biased by an ideal current source $I_1 = 100 \mu\text{A}$ from $V_{DD} = 1.8 \text{ V}$ (V1). Its gate is tied to the gate of NM0, forming the mirror. The source terminals of both devices are grounded. The drain of NM0 is connected to an independent source V0, which is swept for the DC analysis and used to inject a unit AC test signal for the AC analysis.

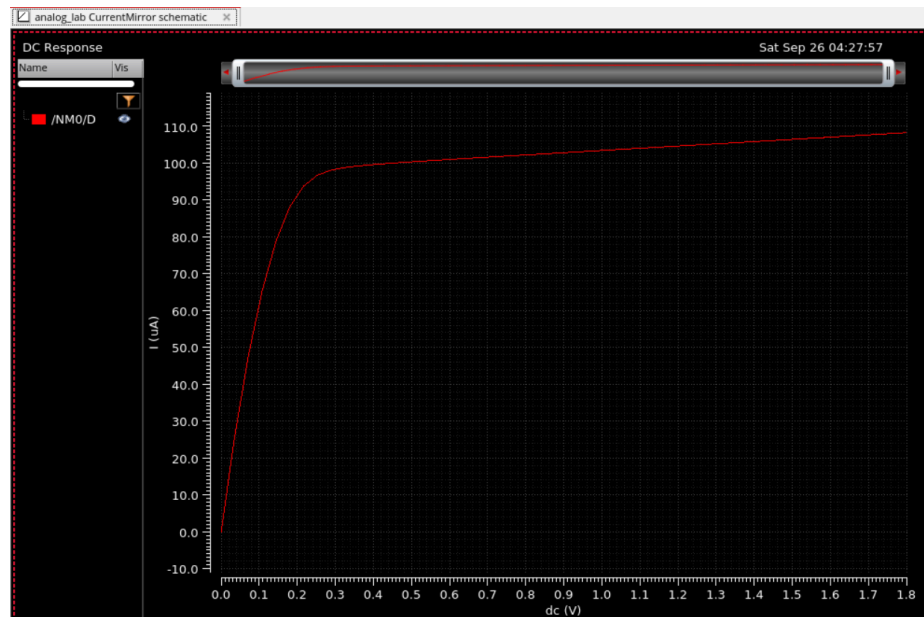


Simulation Procedure

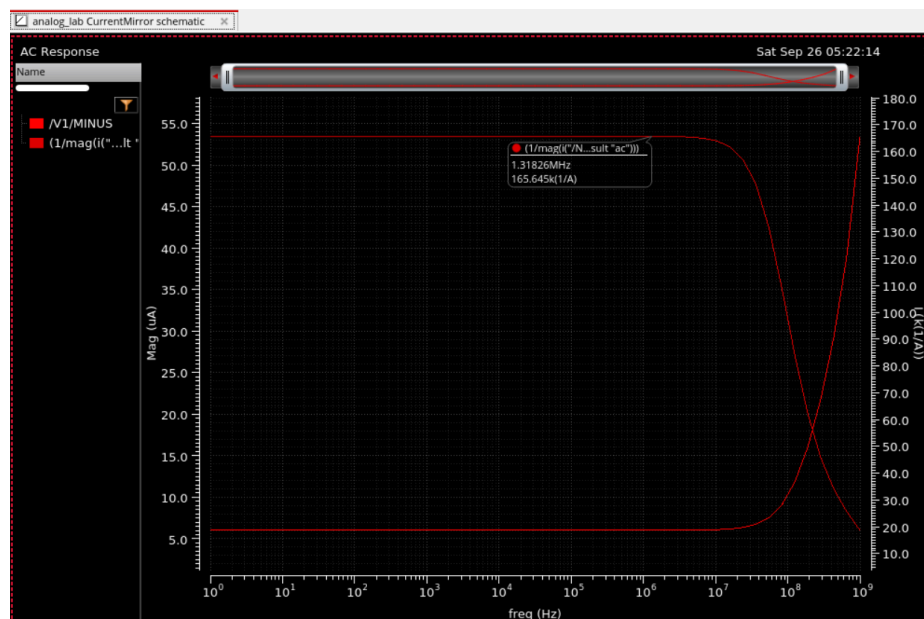
1. DC Analysis: Swept the DC value of V0 from 0 V to 1.8 V, with “Save DC Operating Point” enabled. Plotted the drain current of NM0 (NM0/D) to obtain the I-V characteristic of the output branch.
2. AC Analysis: Swept frequency from 1 Hz to 1 GHz. With V0 supplying a unit AC magnitude at the output node, the AC current through V0 was recorded. The output resistance was computed in the ADE calculator as $R_{out} = 1 / |i(V0)|$ and plotted against frequency.

Results

DC sweep: The output current rises steeply for $V_0 < 0.25$ V and saturates close to $I_{ref} = 100$ μ A once NM0 enters saturation. Beyond this point, the current increases gradually (~ 100 μ A to ~ 108 μ A) as V_0 goes from 0.3 V to 1.8 V, due to channel length modulation.



AC sweep: The output resistance stays flat at approximately 165.6 k Ω from low frequency up to ~ 10 MHz, then rolls off at higher frequencies, with the two traces crossing near 200 MHz.



Observations

- With equal W/L for NM0 and NM1, the mirror ratio is 1, and the output current closely tracks I_{ref} in the saturation region.
- The output resistance is finite ($\sim 165.6 \text{ k}\Omega$), consistent with $r_o = 1/(\lambda \cdot I_D)$.
- The slight upward slope of current with V_0 in saturation confirms the effect of channel length modulation.
- The output resistance falls off at high frequency because parasitic capacitances at the drain node (C_{gd} , C_{db}) provide a low-impedance AC path, degrading the mirror's current-source behavior at RF.

Conclusion

A simple NMOS current mirror was designed and simulated in GPDK090 technology. The DC sweep confirmed correct current mirroring with the output current saturating close to $I_{ref} = 100 \text{ }\mu\text{A}$, and the AC analysis quantified a finite output resistance of $\sim 165.6 \text{ k}\Omega$ that degrades at high frequency due to parasitic capacitance.